

Appendix H — First Hardware Test:

Preregistered FAIL

Phase 1 falsification ledger · Hardware fixed-point test (H1) on IBM Quantum · Run date: 2026-07-11 · Preregistration: prereg_h1_ibm_fixed_point.md (committed to repository before execution)

HEADLINE RESULT: FAIL

Predicted steady-state drift $D^* = 0.0302$ (preregistered PASS window: 0.0272–0.0333). Measured on hardware: $D^* = 0.1463$. Relative deviation: **383.7%**. Kill-condition triggered.

1. What was tested

Hypothesis H1 (preregistered before any hardware run): The Appendix A steady-state drift fixed point holds on a real transmon qubit. Using the device’s published calibration T_2 and the known idle delay per cycle, compute the per-cycle dephasing $\hat{\gamma} = \Delta t/T_2$, and predict the steady-state coherence via the fixed-point map:

$$C^* = f / (1 - (1-f) \cdot e^{-\gamma}) , \quad D^* = (1 - C^*)/2$$

Kill-condition: measured steady-state D^* outside $\pm 10\%$ (relative) of the analytic prediction $\Rightarrow$ FAIL, reported regardless of outcome.

2. Experimental setup

Parameter	Value
Backend	ibm_kingston (Heron r2, 156 qubits, us-east)
Qubit	140 (best calibrated T_2 on device)
Calibration T_2	601.0 μ s
Idle delay per cycle	40 μ s
Correction	Partial re-preparation of $ +\rangle$, gain $f = 0.5$ (probabilistic reset-and-reprepare, classical mixing across circuit families)
Cycles	0–8; steady state = mean of last 3 cycles
Shots	4096 per circuit; 45 circuits, single job
Measurement	X-basis; coherence proxy $C = \langle X \rangle $; drift $D = (1-C)/2$
Job ID	d99brisqp3as739tudkg

3. Prediction on record (Part A, computed before hardware run)

Quantity	Value
γ per cycle (from calibration T2)	0.0665
Per-cycle decay $e^{-\gamma}$	0.9356
Predicted C^*	0.9395
Predicted D^*	0.0302
Preregistered PASS window ($\pm 10\%$)	[0.0272, 0.0333]

4. Hardware results (Part B)

Cycle	C (measured)	$D = (1-C)/2$
0	0.9937	0.0032
1	0.6782	0.1609
2	0.7281	0.1359
3	0.7085	0.1458
4	0.7167	0.1417
5	0.7016	0.1492
6	0.7096	0.1452
7	0.7077	0.1462
8	0.7050	0.1475

Quantity	Predicted	Measured	Deviation	Verdict
Steady-state D^*	0.0302	0.1463	383.7%	FAIL (window $\pm 10\%$)

5. Diagnosis: why it failed

The failure is diagnostic, not random. The single-cycle collapse from $C = 0.994$ to $C = 0.678$ implies an effective per-cycle decay of ≈ 0.68 , versus the 0.936 predicted from calibration T2. The most probable physical cause is a wrong input parameter, not a wrong map:

- **T2 vs T2*:** IBM's published T2 is measured with echo (refocusing) sequences. Our circuit used a *bare idle delay*, which decays at the free-induction time $T2^*$ — typically 5–10 $\times$ shorter due to low-frequency flux/charge noise. This alone accounts for the magnitude of the discrepancy.
- **Structure survived:** the system nonetheless settled into a flat, stable steady state ($C \approx 0.71$ from cycle 2 onward), exactly the qualitative behavior the fixed-point map predicts — at a different level set by the true noise rate.

- **Post-hoc consistency check (not a result):** inserting the *observed* per-cycle decay (≈ 0.68) into the same fixed-point formula gives $C^* \approx 0.758$, $D^* \approx 0.121$, versus measured 0.146. Closer, but obtained after seeing the data; it therefore carries no evidential weight and is recorded only as motivation for H2.

Value of this failure. Appendices A–G were simulations executed by the framework’s authors; hardware is the first arbiter outside our control, and it rejected the prediction as stated. This is precisely what preregistered kill-conditions are for. The lesson is concrete and transferable: simulation cannot catch a wrong noise-parameter assumption (echo-T2 used where free-induction T2* governs). The map’s input model is falsified on bare-idle hardware; the fixed-point structure itself remains untested at the correct noise rate.

6. Sharpened hypothesis H2 (preregisterable; not yet run)

H2: On the same class of device, the identical fixed-point law predicts steady-state drift within the $\pm 10\%$ window when the per-cycle decay parameter is obtained *in situ* — either (a) measured from a short free-induction calibration run immediately before the main experiment, or (b) with echo pulses inserted into the idle delays so that the published echo-T2 becomes the governing rate.

Kill-condition for H2: measured D^* outside $\pm 10\%$ of the prediction computed from the *in situ* rate $\Rightarrow$ the fixed-point law itself (not its inputs) is falsified on hardware.

7. Ledger status after Appendix H

Appendix	Test	Outcome
A	Single-qubit recirculation fixed point (simulation)	PASS (D^* within 2%)
B	Two-qubit inheritance-guided control v1–v2	FAIL (always-B baseline superior)
C–E	Restricted access / crossover law $s^*(y)$	PASS (preregistered predictions confirmed)
F	Optimal-policy gauntlet (control sufficiency)	FAIL (gap 12.19% > 5% threshold)
G	Universality claim	FAIL (“Unified” retired)
H	First hardware test: fixed point with calibration-T2 input (ibm_kingston)	FAIL (383.7% deviation; T2 vs T2* input-model error; H2 defined)

8. Scope statement

Nothing in this appendix claims new physics. The experiment tests whether a standard feedback fixed-point map, with inputs taken from published device calibration, quantitatively predicts hardware behavior. It does not, for the identifiable reason documented above. All code, the preregistration commit, job ID, and raw cycle data are public in the repository for

independent verification. Reported under the Phase 1 rule: every preregistered outcome enters the ledger, pass or fail.

Repository: derekhone/uip-phase1-testbeds · Preregistration: prereg_h1_ibm_fixed_point.md · IBM Quantum open plan · Job d99brisqp3as739tudkg · Generated 2026-07-11